

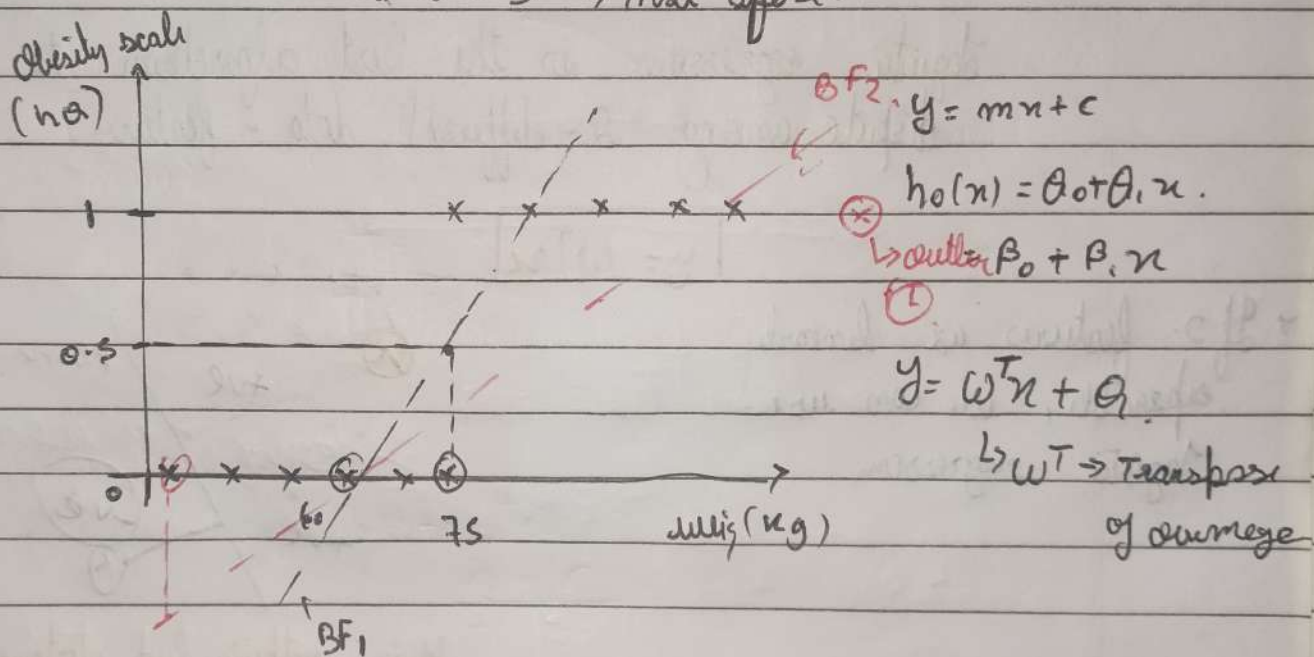
Logistic Regression

① We can't use linear regression always, since, the n (outliers) increases, Best fit lines are to be changed too which causes problems as well as complexity increment

② Say, if a dataset is for class 1 & not class

$$h_0(x) \geq 0.5 \rightarrow \text{class 1}$$

$$\< 0.5 \rightarrow \text{not class 1}$$

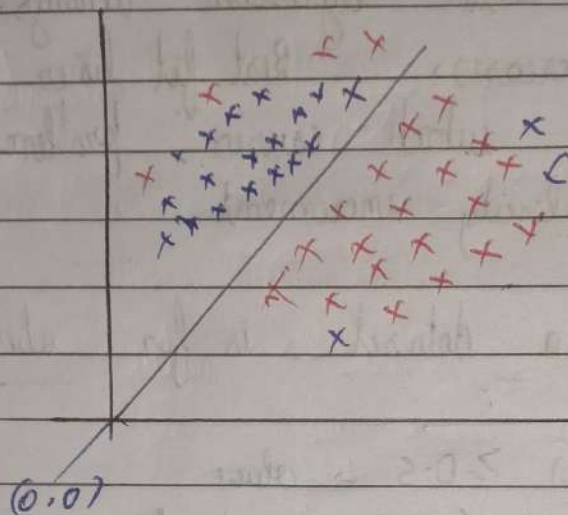


no. given explanation of h_0 to be > 1 or < 0

1
1. ex.

① Geometric Intuition

② Mathematical Intuition



+ve = +1

-ve = -1

$$Eq \Rightarrow y = mx + c$$

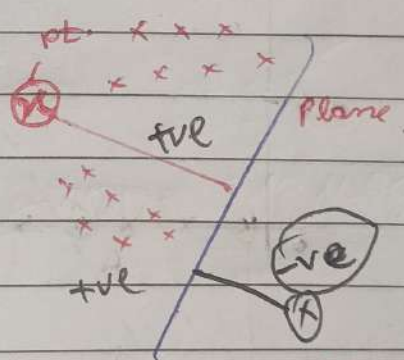
$$y = w^T x + b$$

$b \rightarrow 0$
intercept = 0

Logistic regression is the Best algorithm to dispute among 2 different data - features

$$y = w^T x$$

If 2 features are linearly separable, we can use Logistic regression



y is nothing but distance b/w plane & pt.

$$y = w^T x$$

$$\left\{ \sum_{i=0}^n w_i^T x_i \right\}$$

Case 1: $y = +1$ $w^T x > 0$

$$y = +1$$

$$\omega^T \gamma > 0$$

$$y^T x^{(n)} > 0$$

Case 2:
 $y = -1$, $\omega^T x < 0$

$$y = -1$$

$$\omega^T x < 0$$

$$y^x \omega^T x \geq 0$$

Case 13: $y = -1$, $w^T u > 0$.

$$y^* w^T n \leq 0$$

→ Even

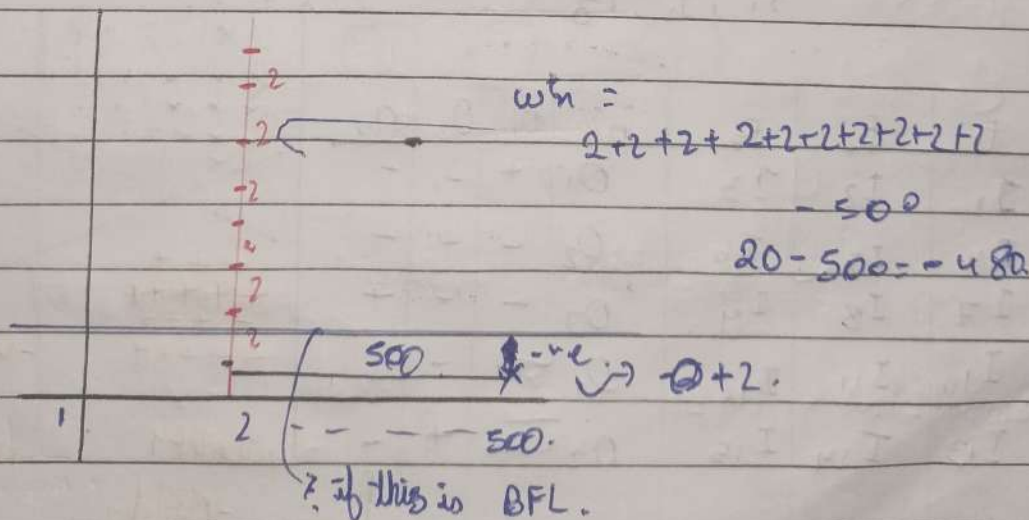
not a good sign for eq.

Thurs 7

$$\text{max} \sum_{i=0}^n y_i w_i x_i$$

← main aim of LR

ii) Example 2 sigmoid function

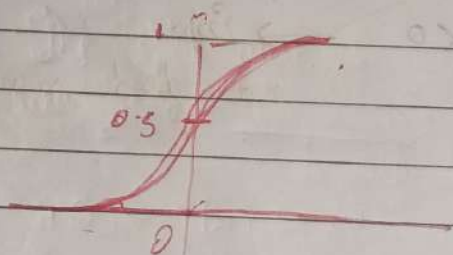


$$\therefore \min \left(\sum_{i=0}^n y_i + w_1^T x_i \right)$$

sigmoid function $\max \left(f \left[\sum_{i=0}^n (y_i \cdot w_i^T x_i) \right] \right)$

$$\left\{ \text{sigmoid} \Rightarrow \frac{1}{1 + e^{-z}} \right\}$$

when -500 is passed through sigmoid
our values are converted
i.e. 0 to 1



(One Vs Rest)

$\rightarrow$ 0vr is used.

day we have more than one feature
 f_1, f_2, f_3

f_1	f_2	f_3	o/p	O_1	O_2	O_3
I_1	I_2	I_3	O_1	+	-	-
I_4	I_5	I_6	O_2	-	+	-
I_7	I_8	I_9	O_3	-	-	+
I_{10}	I_{11}	I_{12}	O_1	+	-	-
I_{13}	I_{14}	I_{15}	O_2	-	+	-

Decision tree Entropy.

Entropy $\rightarrow$ measures the purity of split.

$$H(S) = -P(+) \log_2(P(+)) - P(-) \log_2(P(-))$$

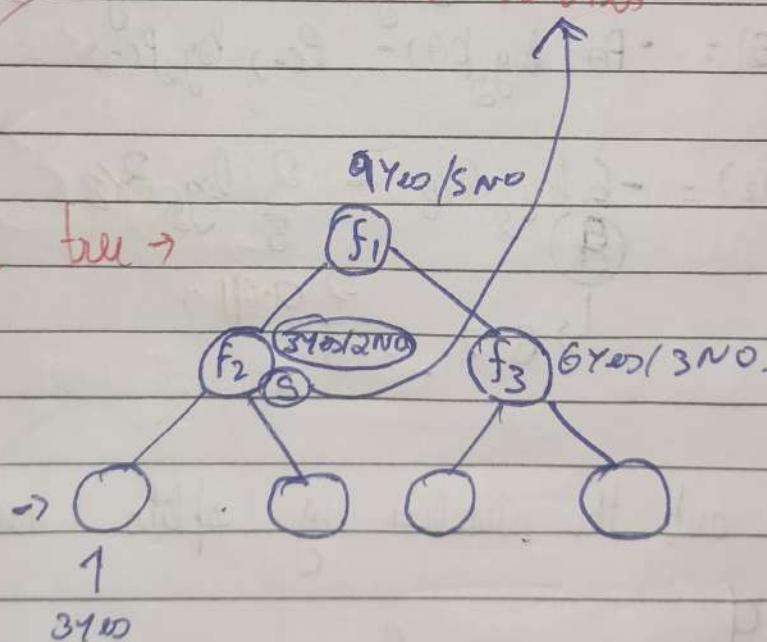
4 Yes / 0 No
 $\rightarrow$ 0 Bit.

$P_+ / P_- = \% \text{ of +ve class} / \% \text{ of -ve class}$

$S = \text{Subset of Train. Ex.}$

$$= -3/5 \log_2(3/5) - (2/5) \log_2(2/5) = 0.92 \text{ bits}$$

from tree $\rightarrow$



8 + 2 = 5
 3 - no. of yes.